

Q.1: Answer the following short questions. (10x3=30 Marks)

(Each question carries 3 marks, answer briefly.)

1. What are the **main responsibilities of a Database Administrator (DBA)**?
 2. Explain **Oracle Instance** and its components.
 3. What is the **role of Data Dictionary in a database**?
 4. Define **Redo Log Files** and explain their significance in database recovery.
 5. What are **Tablespaces**, and how are they managed in Oracle?
 6. Explain **Automatic Undo Management** and its importance.
 7. What are the different **types of Indexes** in Oracle?
 8. Differentiate between **System Privileges and Object Privileges**.
 9. Explain the **role of Oracle Shared Server and Dedicated Server processes**.
 10. What is **Archive Log Mode**, and why is it important for database recovery?
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Q.2: Answer the following long questions. (3x10=30 Marks)

(Each question carries 10 marks, answer in detail.)

1. (a) Explain the **Oracle Architectural Components** with a well-labeled diagram.
(b) How does the **multiplexing of Control Files** enhance database reliability?
2. (a) Discuss the **different Backup and Recovery techniques** used in Oracle.
(b) Explain the step-by-step process of **User-Managed Complete and Incomplete Recovery**.
3. (a) Describe the **Import/Export utility in Oracle**. Explain different export and import modes.
(b) What are **Types of Constraints** in Oracle databases? How are they managed effectively?